

Academic Paper Analysis & Critique

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Cyber Technique: MitM | Evil Twin

Target Paper: A. Kamble and D. Kshirsagar, "Feature Selection in Wireless Intrusion Detection System for Evil Twin Attack Detection," IEEE CISCT 2023.

1. Problem Statement & Literature Gap

Past work achieved accurate Evil Twin detection using heavy models with ~250 features. This presents a speed limitation, which is problematic when the goal of the models is to filter and detect in real time (WIDS) and go through many packets every second. **Core**

Question: Can feature selection techniques filter only relevant features to build light, fast, and effective WIDS models for real-time use?

2. Model Selection & Optimization

- **Feature Selection: Gini Index (GI):** Evaluates and assigns a score (0 to 1) to each feature. Features scoring 0 are eliminated. Outperformed other algorithms (IG, ANOVA, etc.) by reducing the set from 207 to just 25 features, resulting in the lowest build time.
- **Classifier: Random Forest (RF):** A tree-based ensemble classifier. Showed great results in past research and outperformed other tested tree models.
- **Validation:** 10-fold cross-validation implemented via Weka 3.10.

3. Data Pipeline

Flow: [AWID3 Dataset] → [Random Sampling Balancing] → [Preprocessing] → [GI Selection] → [RF Classification]

- **Balancing:** The raw AWID3 dataset was severely imbalanced (3.6M normal vs. 104k attacks). Random sampling reduced this to ~39k normal and ~33k attacks.
- **Preprocessing:** Handled nulls, removed overfit features, and converted strings to numerics, reducing the initial 254 features to 207.
- **Selection:** GI applied to the 207 features. Only 25 non-zero scoring features were kept for classification.

4. Quantitative Results

Evaluated on the AWID3 dataset:

- **Proposed Model (RF + GI):** 99.9945% Accuracy | 99.9898% Recall | Build Time: 11.22s
- **Baseline Model (RF without GI):** 99.9889% Accuracy | 99.9923% Precision | 99.9872% Recall | 5 False Positives | Build Time: 49.35s

5. Scientific Contribution

The main innovation is showing that the attack can be detected with high accuracy using a lean model. By isolating the 25 most critical features out of 254, the authors made the model ~77% faster to set up while maintaining near-perfect recall and precision. The application is extremely practical, as real-time detection requires speed and efficiency.

6. Engineering Takeaways for Project

The 25 final features were derived from the MAC and application layers of the pcap files.

What to Adopt:

- **Strict Preprocessing:** Implement robust cleaning to handle nulls, mixed datatypes, and convert strings to numerics before applying ML models.
- **Data Balancing:** Adopt random sampling to balance skewed network data and prevent the model from becoming biased towards the majority class.
- **Aggressive Feature Selection:** Great results can be achieved with few features, meaning we can process and identify non-relevant features in an aggressive way.

What to Avoid:

- Feeding unfettered data to the model with too many features without testing their relevance.
- **Critical Flaw in Paper:** The researchers ran the feature filtration on all of the data, not just the training data. This is not a practical training procedure (data leakage) and must be avoided.

Proposed Improvement:

1. Prevent data leakage by classifying features of the train only.
2. Test on unbalanced data - while training on unbalanced data is fine testing on it makes the scores twisted.